PHYSIOLOGY

Stomach and Duodenum

SGL 1 · lecturer not printed on slides · GIT BLOCK

Physiology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. What are the three phases of gastric acid secretion, and roughly what % of acid does each contribute?
2. Which stomach region stores food, and which region mixes and empties it?
3. Name two things gastric HCl does besides killing bacteria.

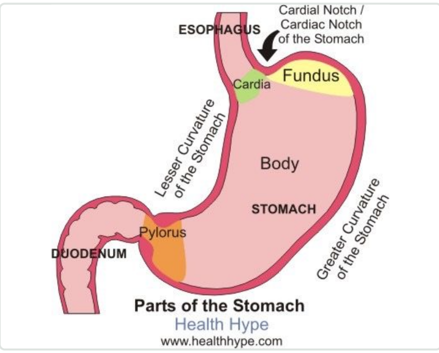
Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Main Functions of the Stomach

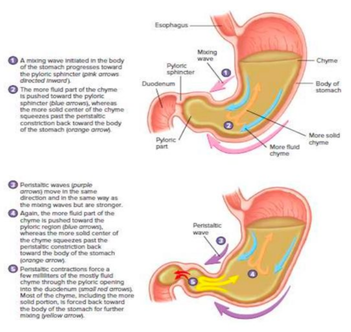
Function	Detail
Storage	Bulk storage of undigested food
Mechanical breakdown	Mixing waves
Chemical breakdown	Acid + pepsin disrupt chemical bonds
Intrinsic factor	Needed for vitamin B12 absorption (later, in the ileum)
Absorption	Very little — but water, some ions, drugs and **alcohol** are absorbed
Endocrine	Enteroendocrine cells produce gastrin

2 Regional anatomy & motility — the key discriminator

Region	Wall	Contraction pattern	Role
Proximal (fundus + body)	Thin	Weak, infrequent — **receptive relaxation**	Stores large volumes of food
Distal (pylorus / antrum)	Thick	Strong, frequent peristalsis	Mixes food and drives **gastric emptying**



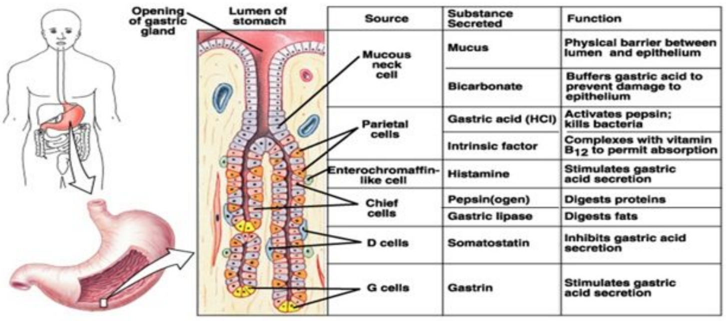
Regions of the stomach — cardia, fundus, body, pylorus.



Mixing and peristaltic waves: fluid chyme is pushed toward the pylorus; solid chyme is squeezed back for further mixing.

3 Gastric secretory cells — who makes what

Cell / source	Secretes	Function
Mucous neck cell	Mucus, bicarbonate	Physical barrier; buffers acid against the epithelium
Parietal cell	HCl, intrinsic factor	Activates pepsin, kills bacteria; IF needed for B12 absorption
ECL cell	Histamine	Directly stimulates parietal cell acid secretion
Chief cell	Pepsinogen, gastric lipase	Digests protein; minor fat digestion
D cell	Somatostatin	Inhibits gastric acid secretion
G cell	Gastrin	Stimulates gastric acid secretion



Gastric gland cell types, their secretions, and functions.

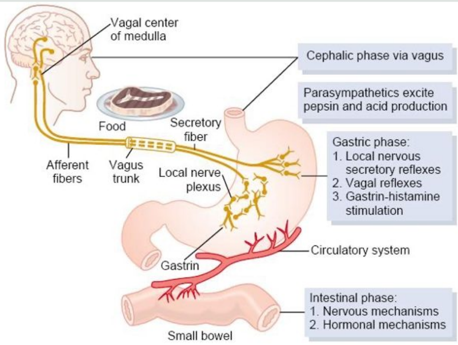
4 Gastric hormones — discriminator grid

> **NEVER MISS:** Somatostatin is the **"master off switch"** — if it's suppressed, acid secretion runs unchecked. Don't confuse it with gastrin, which is the "on switch."

Hormone	Source	Stimulated by	Inhibited by	Key action
Gastrin	G-cells (antrum)	AA/protein, vagal tone, distension, GRP, pH>3, alcohol	pH<3, somatostatin, secretin, CCK, VIP, GIP, glucagon	↑ acid secretion (direct + potentiates histamine); ↑ motility; ↑ mucosal growth
Somatostatin	CNS, antrum, fundus, SI, colon, pancreatic D-cells	Antral acidification; fat/ protein/acid in duodenum; glucose, AA, CCK	ACh (vagal)	Inhibits most GI hormones, secretion, and motility
GRP	Gastric antrum, SI mucosa	Vagal stimulation	—	Stimulates release of **all** GI secretions + motility
Histamine	ECL cells, mast cells	Gastrin, ACh, epinephrine	Somatostatin	Directly stimulates parietal cell acid secretion

5 Phases of stimulated acid secretion — the exam core

Cephalic (20–30%)	Sight, smell, thought or taste of food → cortex & hypothalamus → vagus releases ACh → parietal cell acid production
Gastric (60–70%)	Food enters the stomach → digestion products stimulate G cells → gastrin released; distension alone also raises acid. Lasts until the stomach is empty.
Intestinal (~10%)	Initiated by chyme entering the small bowel; the mechanism is poorly understood



Regulation of gastric secretion across the cephalic, gastric, and intestinal phases.

6 Functions of Gastric HCl

Effect
Activates pepsinogen → pepsin
Provides optimum pH for pepsin action

Denatures protein → helps its digestion
Kills bacteria in food
Helps Fe^{2+} and Ca^{2+} absorption
Promotes pancreatic, small intestinal, and bile secretion

7 The Duodenum — what it does and why it matters

Receives	Chyme from the stomach, plus bile (liver/gallbladder) and pancreatic juice — these aid digestion
Secretes	**Secretin** and **cholecystokinin (CCK)** — both encourage bile and pancreatic juice secretion
Regulates	Rate of gastric emptying; triggers hunger signals
Clinical relevance	Absorbs even more nutrients than the stomach — this is why the duodenum is frequently **bypassed in bariatric surgery** to reduce nutrient absorption in obese patients

P Must-remember pearls

- The **gastric phase** contributes the most acid (60–70%) and lasts as long as food remains in the stomach.
- **Somatostatin** is the master off switch for GI secretion — not gastrin, which is the on switch.
- The stomach barely absorbs nutrients — but it does absorb water, some ions, drugs, and alcohol.
- Proximal stomach (fundus/body) = thin-walled storage; distal stomach (antrum/pylorus) = thick-walled mixing and emptying.
- The **duodenum**, not the stomach, is the segment surgically excluded in gastric bypass to cut nutrient absorption.

Q Test yourself

1. A patient with pernicious anaemia (autoimmune destruction of a gastric cell type) develops B12 deficiency because they've lost:

A. Chief cells
B. Parietal cells (intrinsic factor)
C. G cells
D. Mucous neck cells

2. Which phase of gastric acid secretion contributes the LARGEST share of total acid output?

A. Cephalic phase
B. Gastric phase
C. Intestinal phase
D. All three contribute equally

3. A drug that mimics somatostatin would be expected to:

A. Increase gastric acid secretion
B. Decrease gastric acid secretion
C. Have no effect on acid secretion
D. Selectively increase pepsinogen only

4. A truncal vagotomy (surgically cutting the vagus nerve) would most reduce acid secretion during which phase?

A. Cephalic phase
B. Gastric phase
C. Intestinal phase
D. It would abolish gastric-phase acid entirely

5. In gastric bypass surgery, the duodenum is excluded from the food pathway primarily to reduce:

A. Gastric acid secretion
B. Bile production
C. Nutrient absorption
D. Gastrin release

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — B — parietal cells make both HCl and intrinsic factor; their loss causes achlorhydria and B12 malabsorption.
2 — B — the gastric phase accounts for 60–70% of acid output and continues until the stomach empties.
3 — B — somatostatin is the master inhibitory hormone of GI secretion.
4 — A — the cephalic phase is entirely vagally mediated, so vagotomy abolishes it most directly.
5 — C — the duodenum absorbs more nutrients than the stomach, so bypassing it cuts absorption.

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam